

NCERT CLASS 6

60 Questions Per Topic

Based on the refined 2026–27 curriculum map created for this project.

Each curriculum topic below contains exactly 60 questions: understanding, examples/classification, comparison, application, reasoning/error analysis, and higher-order activity.

These are question-bank items derived from the topic descriptions in the supplied curriculum map. Validate against the exact school/textbook edition before high-stakes use.

MATHEMATICS — 1. Patterns in Mathematics

Coverage: ■ Odd/even; triangular, square and cube patterns; visual patterns; generalising rules.

Understanding

1. What is ■ Odd/even, and what are its main features? Use a simple example.
2. What is triangular, and what are its main features? Use a school-life example.
3. What is square and cube patterns, and what are its main features? Use a real-world example.
4. What is visual patterns, and what are its main features? Show the idea with a diagram where appropriate.
5. What is generalising rules., and what are its main features? Explain in your own words.
6. What is ■ Odd/even, and what are its main features? Give a step-by-step response.
7. What is triangular, and what are its main features? Mention one common misconception.
8. What is square and cube patterns, and what are its main features? State one practical application.
9. What is visual patterns, and what are its main features? Justify your response.
10. What is generalising rules., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Odd/even and explain each. Use a simple example.
12. Give two examples related to triangular and explain each. Use a school-life example.
13. Give two examples related to square and cube patterns and explain each. Use a real-world example.
14. Give two examples related to visual patterns and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to generalising rules. and explain each. Explain in your own words.
16. Give two examples related to ■ Odd/even and explain each. Give a step-by-step response.
17. Give two examples related to triangular and explain each. Mention one common misconception.
18. Give two examples related to square and cube patterns and explain each. State one practical application.
19. Give two examples related to visual patterns and explain each. Justify your response.
20. Give two examples related to generalising rules. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Odd/even related to triangular? Compare the two ideas. Use a simple example.
22. How is triangular related to square and cube patterns? Compare the two ideas. Use a school-life example.
23. How is square and cube patterns related to visual patterns? Compare the two ideas. Use a real-world example.
24. How is visual patterns related to generalising rules.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is generalising rules. related to ■ Odd/even? Compare the two ideas. Explain in your own words.
26. How is ■ Odd/even related to triangular? Compare the two ideas. Give a step-by-step response.
27. How is triangular related to square and cube patterns? Compare the two ideas. Mention one common misconception.
28. How is square and cube patterns related to visual patterns? Compare the two ideas. State one practical application.
29. How is visual patterns related to generalising rules.? Compare the two ideas. Justify your response.
30. How is generalising rules. related to ■ Odd/even? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Odd/even to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply triangular to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply square and cube patterns to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply visual patterns to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply generalising rules. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Odd/even to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply triangular to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply square and cube patterns to a realistic situation. What would you do or calculate? State one practical application.
39. Apply visual patterns to a realistic situation. What would you do or calculate? Justify your response.
40. Apply generalising rules. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Odd/even. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of triangular. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of square and cube patterns. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of visual patterns. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of generalising rules.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Odd/even. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of triangular. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of square and cube patterns. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of visual patterns. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of generalising rules.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Odd/even. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show triangular. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show square and cube patterns. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show visual patterns. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show generalising rules.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Odd/even. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show triangular. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show square and cube patterns. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show visual patterns. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show generalising rules.. Include the relevant rule or principle.

MATHEMATICS — 2. Lines and Angles

Coverage: ■ Points, lines, rays; line segments; angle types; measuring and constructing angles.

Understanding

1. What is ■ Points, and what are its main features? Use a simple example.
2. What is lines, and what are its main features? Use a school-life example.
3. What is rays, and what are its main features? Use a real-world example.
4. What is line segments, and what are its main features? Show the idea with a diagram where appropriate.
5. What is angle types, and what are its main features? Explain in your own words.
6. What is measuring and constructing angles., and what are its main features? Give a step-by-step response.
7. What is ■ Points, and what are its main features? Mention one common misconception.
8. What is lines, and what are its main features? State one practical application.
9. What is rays, and what are its main features? Justify your response.
10. What is line segments, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to angle types and explain each. Use a simple example.
12. Give two examples related to measuring and constructing angles. and explain each. Use a school-life example.
13. Give two examples related to ■ Points and explain each. Use a real-world example.
14. Give two examples related to lines and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to rays and explain each. Explain in your own words.
16. Give two examples related to line segments and explain each. Give a step-by-step response.
17. Give two examples related to angle types and explain each. Mention one common misconception.
18. Give two examples related to measuring and constructing angles. and explain each. State one practical application.
19. Give two examples related to ■ Points and explain each. Justify your response.
20. Give two examples related to lines and explain each. Include the relevant rule or principle.

Comparison

21. How is rays related to line segments? Compare the two ideas. Use a simple example.
22. How is line segments related to angle types? Compare the two ideas. Use a school-life example.
23. How is angle types related to measuring and constructing angles.? Compare the two ideas. Use a real-world example.
24. How is measuring and constructing angles. related to ■ Points? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Points related to lines? Compare the two ideas. Explain in your own words.
26. How is lines related to rays? Compare the two ideas. Give a step-by-step response.
27. How is rays related to line segments? Compare the two ideas. Mention one common misconception.
28. How is line segments related to angle types? Compare the two ideas. State one practical application.
29. How is angle types related to measuring and constructing angles.? Compare the two ideas. Justify your response.
30. How is measuring and constructing angles. related to ■ Points? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Points to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply lines to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply rays to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply line segments to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply angle types to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply measuring and constructing angles. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Points to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply lines to a realistic situation. What would you do or calculate? State one practical application.
39. Apply rays to a realistic situation. What would you do or calculate? Justify your response.
40. Apply line segments to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of angle types. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of measuring and constructing angles.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Points. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of lines. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of rays. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of line segments. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of angle types. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of measuring and constructing angles.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Points. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of lines. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show rays. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show line segments. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show angle types. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show measuring and constructing angles.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Points. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show lines. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show rays. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show line segments. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show angle types. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show measuring and constructing angles.. Include the relevant rule or principle.

MATHEMATICS — 3. Number Play

Coverage: ■ Number puzzles; divisibility ideas; palindromes; Kaprekar-type explorations; reasoning.

Understanding

1. What is ■ Number puzzles, and what are its main features? Use a simple example.
2. What is divisibility ideas, and what are its main features? Use a school-life example.
3. What is palindromes, and what are its main features? Use a real-world example.
4. What is Kaprekar-type explorations, and what are its main features? Show the idea with a diagram where appropriate.
5. What is reasoning., and what are its main features? Explain in your own words.
6. What is ■ Number puzzles, and what are its main features? Give a step-by-step response.
7. What is divisibility ideas, and what are its main features? Mention one common misconception.
8. What is palindromes, and what are its main features? State one practical application.
9. What is Kaprekar-type explorations, and what are its main features? Justify your response.
10. What is reasoning., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Number puzzles and explain each. Use a simple example.
12. Give two examples related to divisibility ideas and explain each. Use a school-life example.
13. Give two examples related to palindromes and explain each. Use a real-world example.
14. Give two examples related to Kaprekar-type explorations and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to reasoning. and explain each. Explain in your own words.
16. Give two examples related to ■ Number puzzles and explain each. Give a step-by-step response.
17. Give two examples related to divisibility ideas and explain each. Mention one common misconception.
18. Give two examples related to palindromes and explain each. State one practical application.
19. Give two examples related to Kaprekar-type explorations and explain each. Justify your response.
20. Give two examples related to reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Number puzzles related to divisibility ideas? Compare the two ideas. Use a simple example.
22. How is divisibility ideas related to palindromes? Compare the two ideas. Use a school-life example.
23. How is palindromes related to Kaprekar-type explorations? Compare the two ideas. Use a real-world example.
24. How is Kaprekar-type explorations related to reasoning.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is reasoning. related to ■ Number puzzles? Compare the two ideas. Explain in your own words.
26. How is ■ Number puzzles related to divisibility ideas? Compare the two ideas. Give a step-by-step response.
27. How is divisibility ideas related to palindromes? Compare the two ideas. Mention one common misconception.
28. How is palindromes related to Kaprekar-type explorations? Compare the two ideas. State one practical application.
29. How is Kaprekar-type explorations related to reasoning.? Compare the two ideas. Justify your response.
30. How is reasoning. related to ■ Number puzzles? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Number puzzles to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply divisibility ideas to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply palindromes to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply Kaprekar-type explorations to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply reasoning. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Number puzzles to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply divisibility ideas to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply palindromes to a realistic situation. What would you do or calculate? State one practical application.
39. Apply Kaprekar-type explorations to a realistic situation. What would you do or calculate? Justify your response.
40. Apply reasoning. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Number puzzles. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of divisibility ideas. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of palindromes. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of Kaprekar-type explorations. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of reasoning.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Number puzzles. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of divisibility ideas. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of palindromes. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of Kaprekar-type explorations. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of reasoning.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Number puzzles. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show divisibility ideas. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show palindromes. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show Kaprekar-type explorations. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show reasoning.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Number puzzles. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show divisibility ideas. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show palindromes. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show Kaprekar-type explorations. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show reasoning.. Include the relevant rule or principle.

MATHEMATICS — 4. Data Handling and Presentation

Coverage: ■ Data collection; tables; tally marks; pictographs; bar graphs; interpretation.

Understanding

1. What is ■ Data collection, and what are its main features? Use a simple example.
2. What is tables, and what are its main features? Use a school-life example.
3. What is tally marks, and what are its main features? Use a real-world example.
4. What is pictographs, and what are its main features? Show the idea with a diagram where appropriate.
5. What is bar graphs, and what are its main features? Explain in your own words.
6. What is interpretation., and what are its main features? Give a step-by-step response.
7. What is ■ Data collection, and what are its main features? Mention one common misconception.
8. What is tables, and what are its main features? State one practical application.
9. What is tally marks, and what are its main features? Justify your response.
10. What is pictographs, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to bar graphs and explain each. Use a simple example.
12. Give two examples related to interpretation. and explain each. Use a school-life example.
13. Give two examples related to ■ Data collection and explain each. Use a real-world example.
14. Give two examples related to tables and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to tally marks and explain each. Explain in your own words.
16. Give two examples related to pictographs and explain each. Give a step-by-step response.
17. Give two examples related to bar graphs and explain each. Mention one common misconception.
18. Give two examples related to interpretation. and explain each. State one practical application.
19. Give two examples related to ■ Data collection and explain each. Justify your response.
20. Give two examples related to tables and explain each. Include the relevant rule or principle.

Comparison

21. How is tally marks related to pictographs? Compare the two ideas. Use a simple example.
22. How is pictographs related to bar graphs? Compare the two ideas. Use a school-life example.
23. How is bar graphs related to interpretation.? Compare the two ideas. Use a real-world example.
24. How is interpretation. related to ■ Data collection? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Data collection related to tables? Compare the two ideas. Explain in your own words.
26. How is tables related to tally marks? Compare the two ideas. Give a step-by-step response.
27. How is tally marks related to pictographs? Compare the two ideas. Mention one common misconception.
28. How is pictographs related to bar graphs? Compare the two ideas. State one practical application.
29. How is bar graphs related to interpretation.? Compare the two ideas. Justify your response.
30. How is interpretation. related to ■ Data collection? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Data collection to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply tables to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply tally marks to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply pictographs to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply bar graphs to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply interpretation. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Data collection to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply tables to a realistic situation. What would you do or calculate? State one practical application.
39. Apply tally marks to a realistic situation. What would you do or calculate? Justify your response.
40. Apply pictographs to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of bar graphs. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of interpretation.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Data collection. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of tables. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of tally marks. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of pictographs. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of bar graphs. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of interpretation.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Data collection. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of tables. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show tally marks. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show pictographs. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show bar graphs. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Data collection. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show tables. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show tally marks. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show pictographs. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show bar graphs. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation.. Include the relevant rule or principle.

MATHEMATICS — 5. Prime Time

Coverage: ■ Factors; multiples; primes/composites; divisibility; HCF and LCM.

Understanding

1. What is ■ Factors, and what are its main features? Use a simple example.
2. What is multiples, and what are its main features? Use a school-life example.
3. What is primes/composites, and what are its main features? Use a real-world example.
4. What is divisibility, and what are its main features? Show the idea with a diagram where appropriate.
5. What is HCF and LCM., and what are its main features? Explain in your own words.
6. What is ■ Factors, and what are its main features? Give a step-by-step response.
7. What is multiples, and what are its main features? Mention one common misconception.
8. What is primes/composites, and what are its main features? State one practical application.
9. What is divisibility, and what are its main features? Justify your response.
10. What is HCF and LCM., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Factors and explain each. Use a simple example.
12. Give two examples related to multiples and explain each. Use a school-life example.
13. Give two examples related to primes/composites and explain each. Use a real-world example.
14. Give two examples related to divisibility and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to HCF and LCM. and explain each. Explain in your own words.
16. Give two examples related to ■ Factors and explain each. Give a step-by-step response.
17. Give two examples related to multiples and explain each. Mention one common misconception.
18. Give two examples related to primes/composites and explain each. State one practical application.
19. Give two examples related to divisibility and explain each. Justify your response.
20. Give two examples related to HCF and LCM. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Factors related to multiples? Compare the two ideas. Use a simple example.
22. How is multiples related to primes/composites? Compare the two ideas. Use a school-life example.
23. How is primes/composites related to divisibility? Compare the two ideas. Use a real-world example.
24. How is divisibility related to HCF and LCM.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is HCF and LCM. related to ■ Factors? Compare the two ideas. Explain in your own words.
26. How is ■ Factors related to multiples? Compare the two ideas. Give a step-by-step response.
27. How is multiples related to primes/composites? Compare the two ideas. Mention one common misconception.
28. How is primes/composites related to divisibility? Compare the two ideas. State one practical application.
29. How is divisibility related to HCF and LCM.? Compare the two ideas. Justify your response.
30. How is HCF and LCM. related to ■ Factors? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Factors to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply multiples to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply primes/composites to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply divisibility to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply HCF and LCM. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Factors to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply multiples to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply primes/composites to a realistic situation. What would you do or calculate? State one practical application.
39. Apply divisibility to a realistic situation. What would you do or calculate? Justify your response.
40. Apply HCF and LCM. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Factors. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of multiples. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of primes/composites. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of divisibility. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of HCF and LCM.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Factors. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of multiples. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of primes/composites. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of divisibility. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of HCF and LCM.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Factors. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show multiples. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show primes/composites. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show divisibility. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show HCF and LCM.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Factors. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show multiples. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show primes/composites. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show divisibility. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show HCF and LCM.. Include the relevant rule or principle.

MATHEMATICS — 6. Perimeter and Area

Coverage: ■ Perimeter/area of rectangles and squares; unit conversion; irregular figures.

Understanding

1. What is ■ Perimeter/area of rectangles and squares, and what are its main features? Use a simple example.
2. What is unit conversion, and what are its main features? Use a school-life example.
3. What is irregular figures., and what are its main features? Use a real-world example.
4. What is ■ Perimeter/area of rectangles and squares, and what are its main features? Show the idea with a diagram where appropriate.
5. What is unit conversion, and what are its main features? Explain in your own words.
6. What is irregular figures., and what are its main features? Give a step-by-step response.
7. What is ■ Perimeter/area of rectangles and squares, and what are its main features? Mention one common misconception.
8. What is unit conversion, and what are its main features? State one practical application.
9. What is irregular figures., and what are its main features? Justify your response.
10. What is ■ Perimeter/area of rectangles and squares, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to unit conversion and explain each. Use a simple example.
12. Give two examples related to irregular figures. and explain each. Use a school-life example.
13. Give two examples related to ■ Perimeter/area of rectangles and squares and explain each. Use a real-world example.
14. Give two examples related to unit conversion and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to irregular figures. and explain each. Explain in your own words.
16. Give two examples related to ■ Perimeter/area of rectangles and squares and explain each. Give a step-by-step response.
17. Give two examples related to unit conversion and explain each. Mention one common misconception.
18. Give two examples related to irregular figures. and explain each. State one practical application.
19. Give two examples related to ■ Perimeter/area of rectangles and squares and explain each. Justify your response.
20. Give two examples related to unit conversion and explain each. Include the relevant rule or principle.

Comparison

21. How is irregular figures. related to ■ Perimeter/area of rectangles and squares? Compare the two ideas. Use a simple example.
22. How is ■ Perimeter/area of rectangles and squares related to unit conversion? Compare the two ideas. Use a school-life example.
23. How is unit conversion related to irregular figures.? Compare the two ideas. Use a real-world example.
24. How is irregular figures. related to ■ Perimeter/area of rectangles and squares? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Perimeter/area of rectangles and squares related to unit conversion? Compare the two ideas. Explain in your own words.
26. How is unit conversion related to irregular figures.? Compare the two ideas. Give a step-by-step response.
27. How is irregular figures. related to ■ Perimeter/area of rectangles and squares? Compare the two ideas. Mention one common misconception.
28. How is ■ Perimeter/area of rectangles and squares related to unit conversion? Compare the two ideas. State one practical application.
29. How is unit conversion related to irregular figures.? Compare the two ideas. Justify your response.
30. How is irregular figures. related to ■ Perimeter/area of rectangles and squares? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Perimeter/area of rectangles and squares to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply unit conversion to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply irregular figures. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Perimeter/area of rectangles and squares to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply unit conversion to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply irregular figures. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Perimeter/area of rectangles and squares to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply unit conversion to a realistic situation. What would you do or calculate? State one practical application.
39. Apply irregular figures. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Perimeter/area of rectangles and squares to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of unit conversion. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of irregular figures.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Perimeter/area of rectangles and squares. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of unit conversion. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of irregular figures.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Perimeter/area of rectangles and squares. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of unit conversion. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of irregular figures.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Perimeter/area of rectangles and squares. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of unit conversion. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show irregular figures.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Perimeter/area of rectangles and squares. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show unit conversion. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show irregular figures.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Perimeter/area of rectangles and squares. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show unit conversion. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show irregular figures.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Perimeter/area of rectangles and squares. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show unit conversion. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show irregular figures.. Include the relevant rule or principle.

MATHEMATICS — 7. Fractions

Coverage: ■ Equivalent fractions; comparison/order; addition/subtraction; multiplication/division concepts.

Understanding

1. What is ■ Equivalent fractions, and what are its main features? Use a simple example.
2. What is comparison/order, and what are its main features? Use a school-life example.
3. What is addition/subtraction, and what are its main features? Use a real-world example.
4. What is multiplication/division concepts., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Equivalent fractions, and what are its main features? Explain in your own words.
6. What is comparison/order, and what are its main features? Give a step-by-step response.
7. What is addition/subtraction, and what are its main features? Mention one common misconception.
8. What is multiplication/division concepts., and what are its main features? State one practical application.
9. What is ■ Equivalent fractions, and what are its main features? Justify your response.
10. What is comparison/order, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to addition/subtraction and explain each. Use a simple example.
12. Give two examples related to multiplication/division concepts. and explain each. Use a school-life example.
13. Give two examples related to ■ Equivalent fractions and explain each. Use a real-world example.
14. Give two examples related to comparison/order and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to addition/subtraction and explain each. Explain in your own words.
16. Give two examples related to multiplication/division concepts. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Equivalent fractions and explain each. Mention one common misconception.
18. Give two examples related to comparison/order and explain each. State one practical application.
19. Give two examples related to addition/subtraction and explain each. Justify your response.
20. Give two examples related to multiplication/division concepts. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Equivalent fractions related to comparison/order? Compare the two ideas. Use a simple example.
22. How is comparison/order related to addition/subtraction? Compare the two ideas. Use a school-life example.
23. How is addition/subtraction related to multiplication/division concepts.? Compare the two ideas. Use a real-world example.
24. How is multiplication/division concepts. related to ■ Equivalent fractions? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Equivalent fractions related to comparison/order? Compare the two ideas. Explain in your own words.
26. How is comparison/order related to addition/subtraction? Compare the two ideas. Give a step-by-step response.
27. How is addition/subtraction related to multiplication/division concepts.? Compare the two ideas. Mention one common misconception.
28. How is multiplication/division concepts. related to ■ Equivalent fractions? Compare the two ideas. State one practical application.
29. How is ■ Equivalent fractions related to comparison/order? Compare the two ideas. Justify your response.
30. How is comparison/order related to addition/subtraction? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply addition/subtraction to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply multiplication/division concepts. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Equivalent fractions to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply comparison/order to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply addition/subtraction to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply multiplication/division concepts. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Equivalent fractions to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply comparison/order to a realistic situation. What would you do or calculate? State one practical application.
39. Apply addition/subtraction to a realistic situation. What would you do or calculate? Justify your response.
40. Apply multiplication/division concepts. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Equivalent fractions. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of comparison/order. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of addition/subtraction. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of multiplication/division concepts.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Equivalent fractions. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of comparison/order. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of addition/subtraction. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of multiplication/division concepts.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Equivalent fractions. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of comparison/order. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show addition/subtraction. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show multiplication/division concepts.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Equivalent fractions. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show comparison/order. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show addition/subtraction. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show multiplication/division concepts.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Equivalent fractions. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show comparison/order. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show addition/subtraction. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show multiplication/division concepts.. Include the relevant rule or principle.

MATHEMATICS — 8. Playing with Constructions

Coverage: ■ Ruler-compass constructions; circles; perpendiculars; angle constructions.

Understanding

1. What is ■ Ruler-compass constructions, and what are its main features? Use a simple example.
2. What is circles, and what are its main features? Use a school-life example.
3. What is perpendiculars, and what are its main features? Use a real-world example.
4. What is angle constructions., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Ruler-compass constructions, and what are its main features? Explain in your own words.
6. What is circles, and what are its main features? Give a step-by-step response.
7. What is perpendiculars, and what are its main features? Mention one common misconception.
8. What is angle constructions., and what are its main features? State one practical application.
9. What is ■ Ruler-compass constructions, and what are its main features? Justify your response.
10. What is circles, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to perpendiculars and explain each. Use a simple example.
12. Give two examples related to angle constructions. and explain each. Use a school-life example.
13. Give two examples related to ■ Ruler-compass constructions and explain each. Use a real-world example.
14. Give two examples related to circles and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to perpendiculars and explain each. Explain in your own words.
16. Give two examples related to angle constructions. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Ruler-compass constructions and explain each. Mention one common misconception.
18. Give two examples related to circles and explain each. State one practical application.
19. Give two examples related to perpendiculars and explain each. Justify your response.
20. Give two examples related to angle constructions. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Ruler-compass constructions related to circles? Compare the two ideas. Use a simple example.
22. How is circles related to perpendiculars? Compare the two ideas. Use a school-life example.
23. How is perpendiculars related to angle constructions.? Compare the two ideas. Use a real-world example.
24. How is angle constructions. related to ■ Ruler-compass constructions? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Ruler-compass constructions related to circles? Compare the two ideas. Explain in your own words.
26. How is circles related to perpendiculars? Compare the two ideas. Give a step-by-step response.
27. How is perpendiculars related to angle constructions.? Compare the two ideas. Mention one common misconception.
28. How is angle constructions. related to ■ Ruler-compass constructions? Compare the two ideas. State one practical application.
29. How is ■ Ruler-compass constructions related to circles? Compare the two ideas. Justify your response.
30. How is circles related to perpendiculars? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply perpendiculars to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply angle constructions. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Ruler-compass constructions to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply circles to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply perpendiculars to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply angle constructions. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Ruler-compass constructions to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply circles to a realistic situation. What would you do or calculate? State one practical application.
39. Apply perpendiculars to a realistic situation. What would you do or calculate? Justify your response.
40. Apply angle constructions. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Ruler-compass constructions. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of circles. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of perpendiculars. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of angle constructions.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Ruler-compass constructions. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of circles. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of perpendiculars. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of angle constructions.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Ruler-compass constructions. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of circles. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show perpendiculars. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show angle constructions.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ruler-compass constructions. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show circles. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show perpendiculars. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show angle constructions.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ruler-compass constructions. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show circles. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show perpendiculars. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show angle constructions.. Include the relevant rule or principle.

MATHEMATICS — 9. Symmetry

Coverage: ■ Line and rotational symmetry; symmetry in nature, art and patterns.

Understanding

1. What is ■ Line and rotational symmetry, and what are its main features? Use a simple example.
2. What is symmetry in nature, and what are its main features? Use a school-life example.
3. What is art and patterns., and what are its main features? Use a real-world example.
4. What is ■ Line and rotational symmetry, and what are its main features? Show the idea with a diagram where appropriate.
5. What is symmetry in nature, and what are its main features? Explain in your own words.
6. What is art and patterns., and what are its main features? Give a step-by-step response.
7. What is ■ Line and rotational symmetry, and what are its main features? Mention one common misconception.
8. What is symmetry in nature, and what are its main features? State one practical application.
9. What is art and patterns., and what are its main features? Justify your response.
10. What is ■ Line and rotational symmetry, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to symmetry in nature and explain each. Use a simple example.
12. Give two examples related to art and patterns. and explain each. Use a school-life example.
13. Give two examples related to ■ Line and rotational symmetry and explain each. Use a real-world example.
14. Give two examples related to symmetry in nature and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to art and patterns. and explain each. Explain in your own words.
16. Give two examples related to ■ Line and rotational symmetry and explain each. Give a step-by-step response.
17. Give two examples related to symmetry in nature and explain each. Mention one common misconception.
18. Give two examples related to art and patterns. and explain each. State one practical application.
19. Give two examples related to ■ Line and rotational symmetry and explain each. Justify your response.
20. Give two examples related to symmetry in nature and explain each. Include the relevant rule or principle.

Comparison

21. How is art and patterns. related to ■ Line and rotational symmetry? Compare the two ideas. Use a simple example.
22. How is ■ Line and rotational symmetry related to symmetry in nature? Compare the two ideas. Use a school-life example.
23. How is symmetry in nature related to art and patterns.? Compare the two ideas. Use a real-world example.
24. How is art and patterns. related to ■ Line and rotational symmetry? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Line and rotational symmetry related to symmetry in nature? Compare the two ideas. Explain in your own words.
26. How is symmetry in nature related to art and patterns.? Compare the two ideas. Give a step-by-step response.
27. How is art and patterns. related to ■ Line and rotational symmetry? Compare the two ideas. Mention one common misconception.
28. How is ■ Line and rotational symmetry related to symmetry in nature? Compare the two ideas. State one practical application.
29. How is symmetry in nature related to art and patterns.? Compare the two ideas. Justify your response.
30. How is art and patterns. related to ■ Line and rotational symmetry? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Line and rotational symmetry to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply symmetry in nature to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply art and patterns. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Line and rotational symmetry to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply symmetry in nature to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply art and patterns. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Line and rotational symmetry to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply symmetry in nature to a realistic situation. What would you do or calculate? State one practical application.
39. Apply art and patterns. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Line and rotational symmetry to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of symmetry in nature. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of art and patterns.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Line and rotational symmetry. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of symmetry in nature. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of art and patterns.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Line and rotational symmetry. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of symmetry in nature. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of art and patterns.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Line and rotational symmetry. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of symmetry in nature. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show art and patterns.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line and rotational symmetry. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry in nature. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show art and patterns.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line and rotational symmetry. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry in nature. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show art and patterns.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line and rotational symmetry. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry in nature. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show art and patterns.. Include the relevant rule or principle.

MATHEMATICS — 10. The Other Side of Zero

Coverage: ■ Integers; number line; comparison; addition/subtraction; real-life negative quantities.

Understanding

1. What is ■ Integers, and what are its main features? Use a simple example.
2. What is number line, and what are its main features? Use a school-life example.
3. What is comparison, and what are its main features? Use a real-world example.
4. What is addition/subtraction, and what are its main features? Show the idea with a diagram where appropriate.
5. What is real-life negative quantities., and what are its main features? Explain in your own words.
6. What is ■ Integers, and what are its main features? Give a step-by-step response.
7. What is number line, and what are its main features? Mention one common misconception.
8. What is comparison, and what are its main features? State one practical application.
9. What is addition/subtraction, and what are its main features? Justify your response.
10. What is real-life negative quantities., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Integers and explain each. Use a simple example.
12. Give two examples related to number line and explain each. Use a school-life example.
13. Give two examples related to comparison and explain each. Use a real-world example.
14. Give two examples related to addition/subtraction and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to real-life negative quantities. and explain each. Explain in your own words.
16. Give two examples related to ■ Integers and explain each. Give a step-by-step response.
17. Give two examples related to number line and explain each. Mention one common misconception.
18. Give two examples related to comparison and explain each. State one practical application.
19. Give two examples related to addition/subtraction and explain each. Justify your response.
20. Give two examples related to real-life negative quantities. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Integers related to number line? Compare the two ideas. Use a simple example.
22. How is number line related to comparison? Compare the two ideas. Use a school-life example.
23. How is comparison related to addition/subtraction? Compare the two ideas. Use a real-world example.
24. How is addition/subtraction related to real-life negative quantities.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is real-life negative quantities. related to ■ Integers? Compare the two ideas. Explain in your own words.
26. How is ■ Integers related to number line? Compare the two ideas. Give a step-by-step response.
27. How is number line related to comparison? Compare the two ideas. Mention one common misconception.
28. How is comparison related to addition/subtraction? Compare the two ideas. State one practical application.
29. How is addition/subtraction related to real-life negative quantities.? Compare the two ideas. Justify your response.
30. How is real-life negative quantities. related to ■ Integers? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Integers to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply number line to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply comparison to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply addition/subtraction to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply real-life negative quantities. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Integers to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply number line to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply comparison to a realistic situation. What would you do or calculate? State one practical application.
39. Apply addition/subtraction to a realistic situation. What would you do or calculate? Justify your response.
40. Apply real-life negative quantities. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Integers. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of number line. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of comparison. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of addition/subtraction. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of real-life negative quantities.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Integers. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of number line. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of comparison. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of addition/subtraction. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of real-life negative quantities.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Integers. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show number line. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show comparison. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show addition/subtraction. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show real-life negative quantities.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Integers. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show number line. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show comparison. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show addition/subtraction. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show real-life negative quantities.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 1. The Wonderful World of Science

Coverage: ■ Scientific inquiry; observation; questions; evidence; branches and applications of science.

Understanding

1. What is ■ Scientific inquiry, and what are its main features? Use a simple example.
2. What is observation, and what are its main features? Use a school-life example.
3. What is questions, and what are its main features? Use a real-world example.
4. What is evidence, and what are its main features? Show the idea with a diagram where appropriate.
5. What is branches and applications of science., and what are its main features? Explain in your own words.
6. What is ■ Scientific inquiry, and what are its main features? Give a step-by-step response.
7. What is observation, and what are its main features? Mention one common misconception.
8. What is questions, and what are its main features? State one practical application.
9. What is evidence, and what are its main features? Justify your response.
10. What is branches and applications of science., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Scientific inquiry and explain each. Use a simple example.
12. Give two examples related to observation and explain each. Use a school-life example.
13. Give two examples related to questions and explain each. Use a real-world example.
14. Give two examples related to evidence and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to branches and applications of science. and explain each. Explain in your own words.
16. Give two examples related to ■ Scientific inquiry and explain each. Give a step-by-step response.
17. Give two examples related to observation and explain each. Mention one common misconception.
18. Give two examples related to questions and explain each. State one practical application.
19. Give two examples related to evidence and explain each. Justify your response.
20. Give two examples related to branches and applications of science. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Scientific inquiry related to observation? Compare the two ideas. Use a simple example.
22. How is observation related to questions? Compare the two ideas. Use a school-life example.
23. How is questions related to evidence? Compare the two ideas. Use a real-world example.
24. How is evidence related to branches and applications of science.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is branches and applications of science. related to ■ Scientific inquiry? Compare the two ideas. Explain in your own words.
26. How is ■ Scientific inquiry related to observation? Compare the two ideas. Give a step-by-step response.
27. How is observation related to questions? Compare the two ideas. Mention one common misconception.
28. How is questions related to evidence? Compare the two ideas. State one practical application.
29. How is evidence related to branches and applications of science.? Compare the two ideas. Justify your response.
30. How is branches and applications of science. related to ■ Scientific inquiry? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Scientific inquiry to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply observation to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply questions to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply evidence to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply branches and applications of science. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Scientific inquiry to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply observation to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply questions to a realistic situation. What would you do or calculate? State one practical application.
39. Apply evidence to a realistic situation. What would you do or calculate? Justify your response.
40. Apply branches and applications of science. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Scientific inquiry. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of observation. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of questions. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of evidence. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of branches and applications of science.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Scientific inquiry. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of observation. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of questions. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of evidence. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of branches and applications of science.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Scientific inquiry. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show observation. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show questions. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show evidence. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show branches and applications of science.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Scientific inquiry. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show observation. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show questions. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show evidence. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show branches and applications of science.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 2. Diversity in the Living World

Coverage: ■ Characteristics of living things; classification; habitats; biodiversity.

Understanding

1. What is ■ Characteristics of living things, and what are its main features? Use a simple example.
2. What is classification, and what are its main features? Use a school-life example.
3. What is habitats, and what are its main features? Use a real-world example.
4. What is biodiversity., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Characteristics of living things, and what are its main features? Explain in your own words.
6. What is classification, and what are its main features? Give a step-by-step response.
7. What is habitats, and what are its main features? Mention one common misconception.
8. What is biodiversity., and what are its main features? State one practical application.
9. What is ■ Characteristics of living things, and what are its main features? Justify your response.
10. What is classification, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to habitats and explain each. Use a simple example.
12. Give two examples related to biodiversity. and explain each. Use a school-life example.
13. Give two examples related to ■ Characteristics of living things and explain each. Use a real-world example.
14. Give two examples related to classification and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to habitats and explain each. Explain in your own words.
16. Give two examples related to biodiversity. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Characteristics of living things and explain each. Mention one common misconception.
18. Give two examples related to classification and explain each. State one practical application.
19. Give two examples related to habitats and explain each. Justify your response.
20. Give two examples related to biodiversity. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Characteristics of living things related to classification? Compare the two ideas. Use a simple example.
22. How is classification related to habitats? Compare the two ideas. Use a school-life example.
23. How is habitats related to biodiversity.? Compare the two ideas. Use a real-world example.
24. How is biodiversity. related to ■ Characteristics of living things? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Characteristics of living things related to classification? Compare the two ideas. Explain in your own words.
26. How is classification related to habitats? Compare the two ideas. Give a step-by-step response.
27. How is habitats related to biodiversity.? Compare the two ideas. Mention one common misconception.
28. How is biodiversity. related to ■ Characteristics of living things? Compare the two ideas. State one practical application.
29. How is ■ Characteristics of living things related to classification? Compare the two ideas. Justify your response.
30. How is classification related to habitats? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply habitats to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply biodiversity. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Characteristics of living things to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply classification to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply habitats to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply biodiversity. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Characteristics of living things to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply classification to a realistic situation. What would you do or calculate? State one practical application.
39. Apply habitats to a realistic situation. What would you do or calculate? Justify your response.
40. Apply biodiversity. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Characteristics of living things. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of classification. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of habitats. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of biodiversity.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Characteristics of living things. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of classification. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of habitats. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of biodiversity.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Characteristics of living things. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of classification. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show habitats. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show biodiversity.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Characteristics of living things. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show classification. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show habitats. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show biodiversity.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Characteristics of living things. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show classification. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show habitats. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show biodiversity.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 3. Mindful Eating: A Path to a Healthy Body

Coverage: ■ Nutrients; balanced diet; deficiency; food choices; health.

Understanding

1. What is ■ Nutrients, and what are its main features? Use a simple example.
2. What is balanced diet, and what are its main features? Use a school-life example.
3. What is deficiency, and what are its main features? Use a real-world example.
4. What is food choices, and what are its main features? Show the idea with a diagram where appropriate.
5. What is health., and what are its main features? Explain in your own words.
6. What is ■ Nutrients, and what are its main features? Give a step-by-step response.
7. What is balanced diet, and what are its main features? Mention one common misconception.
8. What is deficiency, and what are its main features? State one practical application.
9. What is food choices, and what are its main features? Justify your response.
10. What is health., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Nutrients and explain each. Use a simple example.
12. Give two examples related to balanced diet and explain each. Use a school-life example.
13. Give two examples related to deficiency and explain each. Use a real-world example.
14. Give two examples related to food choices and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to health. and explain each. Explain in your own words.
16. Give two examples related to ■ Nutrients and explain each. Give a step-by-step response.
17. Give two examples related to balanced diet and explain each. Mention one common misconception.
18. Give two examples related to deficiency and explain each. State one practical application.
19. Give two examples related to food choices and explain each. Justify your response.
20. Give two examples related to health. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Nutrients related to balanced diet? Compare the two ideas. Use a simple example.
22. How is balanced diet related to deficiency? Compare the two ideas. Use a school-life example.
23. How is deficiency related to food choices? Compare the two ideas. Use a real-world example.
24. How is food choices related to health.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is health. related to ■ Nutrients? Compare the two ideas. Explain in your own words.
26. How is ■ Nutrients related to balanced diet? Compare the two ideas. Give a step-by-step response.
27. How is balanced diet related to deficiency? Compare the two ideas. Mention one common misconception.
28. How is deficiency related to food choices? Compare the two ideas. State one practical application.
29. How is food choices related to health.? Compare the two ideas. Justify your response.
30. How is health. related to ■ Nutrients? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Nutrients to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply balanced diet to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply deficiency to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply food choices to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply health. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Nutrients to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply balanced diet to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply deficiency to a realistic situation. What would you do or calculate? State one practical application.
39. Apply food choices to a realistic situation. What would you do or calculate? Justify your response.
40. Apply health. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Nutrients. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of balanced diet. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of deficiency. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of food choices. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of health.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Nutrients. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of balanced diet. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of deficiency. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of food choices. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of health.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nutrients. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show balanced diet. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show deficiency. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show food choices. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show health.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Nutrients. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show balanced diet. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show deficiency. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show food choices. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show health.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 4. Exploring Magnets

Coverage: ■ Magnetic/non-magnetic materials; poles; attraction/repulsion; compass.

Understanding

1. What is ■ Magnetic/non-magnetic materials, and what are its main features? Use a simple example.
2. What is poles, and what are its main features? Use a school-life example.
3. What is attraction/repulsion, and what are its main features? Use a real-world example.
4. What is compass., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Magnetic/non-magnetic materials, and what are its main features? Explain in your own words.
6. What is poles, and what are its main features? Give a step-by-step response.
7. What is attraction/repulsion, and what are its main features? Mention one common misconception.
8. What is compass., and what are its main features? State one practical application.
9. What is ■ Magnetic/non-magnetic materials, and what are its main features? Justify your response.
10. What is poles, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to attraction/repulsion and explain each. Use a simple example.
12. Give two examples related to compass. and explain each. Use a school-life example.
13. Give two examples related to ■ Magnetic/non-magnetic materials and explain each. Use a real-world example.
14. Give two examples related to poles and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to attraction/repulsion and explain each. Explain in your own words.
16. Give two examples related to compass. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Magnetic/non-magnetic materials and explain each. Mention one common misconception.
18. Give two examples related to poles and explain each. State one practical application.
19. Give two examples related to attraction/repulsion and explain each. Justify your response.
20. Give two examples related to compass. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Magnetic/non-magnetic materials related to poles? Compare the two ideas. Use a simple example.
22. How is poles related to attraction/repulsion? Compare the two ideas. Use a school-life example.
23. How is attraction/repulsion related to compass.? Compare the two ideas. Use a real-world example.
24. How is compass. related to ■ Magnetic/non-magnetic materials? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Magnetic/non-magnetic materials related to poles? Compare the two ideas. Explain in your own words.
26. How is poles related to attraction/repulsion? Compare the two ideas. Give a step-by-step response.
27. How is attraction/repulsion related to compass.? Compare the two ideas. Mention one common misconception.
28. How is compass. related to ■ Magnetic/non-magnetic materials? Compare the two ideas. State one practical application.
29. How is ■ Magnetic/non-magnetic materials related to poles? Compare the two ideas. Justify your response.
30. How is poles related to attraction/repulsion? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply attraction/repulsion to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply compass. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Magnetic/non-magnetic materials to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply poles to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply attraction/repulsion to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply compass. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Magnetic/non-magnetic materials to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply poles to a realistic situation. What would you do or calculate? State one practical application.
39. Apply attraction/repulsion to a realistic situation. What would you do or calculate? Justify your response.
40. Apply compass. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Magnetic/non-magnetic materials. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of poles. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of attraction/repulsion. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of compass.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Magnetic/non-magnetic materials. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of poles. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of attraction/repulsion. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of compass.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Magnetic/non-magnetic materials. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of poles. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show attraction/repulsion. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show compass.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Magnetic/non-magnetic materials. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show poles. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show attraction/repulsion. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show compass.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Magnetic/non-magnetic materials. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show poles. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show attraction/repulsion. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show compass.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 5. Measurement of Length and Motion

Coverage: ■ Units; measuring accurately; reference points; types of motion.

Understanding

1. What is ■ Units, and what are its main features? Use a simple example.
2. What is measuring accurately, and what are its main features? Use a school-life example.
3. What is reference points, and what are its main features? Use a real-world example.
4. What is types of motion., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Units, and what are its main features? Explain in your own words.
6. What is measuring accurately, and what are its main features? Give a step-by-step response.
7. What is reference points, and what are its main features? Mention one common misconception.
8. What is types of motion., and what are its main features? State one practical application.
9. What is ■ Units, and what are its main features? Justify your response.
10. What is measuring accurately, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to reference points and explain each. Use a simple example.
12. Give two examples related to types of motion. and explain each. Use a school-life example.
13. Give two examples related to ■ Units and explain each. Use a real-world example.
14. Give two examples related to measuring accurately and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to reference points and explain each. Explain in your own words.
16. Give two examples related to types of motion. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Units and explain each. Mention one common misconception.
18. Give two examples related to measuring accurately and explain each. State one practical application.
19. Give two examples related to reference points and explain each. Justify your response.
20. Give two examples related to types of motion. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Units related to measuring accurately? Compare the two ideas. Use a simple example.
22. How is measuring accurately related to reference points? Compare the two ideas. Use a school-life example.
23. How is reference points related to types of motion.? Compare the two ideas. Use a real-world example.
24. How is types of motion. related to ■ Units? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Units related to measuring accurately? Compare the two ideas. Explain in your own words.
26. How is measuring accurately related to reference points? Compare the two ideas. Give a step-by-step response.
27. How is reference points related to types of motion.? Compare the two ideas. Mention one common misconception.
28. How is types of motion. related to ■ Units? Compare the two ideas. State one practical application.
29. How is ■ Units related to measuring accurately? Compare the two ideas. Justify your response.
30. How is measuring accurately related to reference points? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply reference points to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply types of motion. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Units to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply measuring accurately to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply reference points to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply types of motion. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Units to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply measuring accurately to a realistic situation. What would you do or calculate? State one practical application.
39. Apply reference points to a realistic situation. What would you do or calculate? Justify your response.
40. Apply types of motion. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Units. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of measuring accurately. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of reference points. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of types of motion.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Units. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of measuring accurately. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of reference points. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of types of motion.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Units. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of measuring accurately. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show reference points. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show types of motion.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Units. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show measuring accurately. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show reference points. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show types of motion.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Units. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show measuring accurately. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show reference points. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show types of motion.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 6. Materials Around Us

Coverage: ■ Properties and uses of materials; solids/liquids/gases; classification.

Understanding

1. What is ■ Properties and uses of materials, and what are its main features? Use a simple example.
2. What is solids/liquids/gases, and what are its main features? Use a school-life example.
3. What is classification., and what are its main features? Use a real-world example.
4. What is ■ Properties and uses of materials, and what are its main features? Show the idea with a diagram where appropriate.
5. What is solids/liquids/gases, and what are its main features? Explain in your own words.
6. What is classification., and what are its main features? Give a step-by-step response.
7. What is ■ Properties and uses of materials, and what are its main features? Mention one common misconception.
8. What is solids/liquids/gases, and what are its main features? State one practical application.
9. What is classification., and what are its main features? Justify your response.
10. What is ■ Properties and uses of materials, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to solids/liquids/gases and explain each. Use a simple example.
12. Give two examples related to classification. and explain each. Use a school-life example.
13. Give two examples related to ■ Properties and uses of materials and explain each. Use a real-world example.
14. Give two examples related to solids/liquids/gases and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to classification. and explain each. Explain in your own words.
16. Give two examples related to ■ Properties and uses of materials and explain each. Give a step-by-step response.
17. Give two examples related to solids/liquids/gases and explain each. Mention one common misconception.
18. Give two examples related to classification. and explain each. State one practical application.
19. Give two examples related to ■ Properties and uses of materials and explain each. Justify your response.
20. Give two examples related to solids/liquids/gases and explain each. Include the relevant rule or principle.

Comparison

21. How is classification. related to ■ Properties and uses of materials? Compare the two ideas. Use a simple example.
22. How is ■ Properties and uses of materials related to solids/liquids/gases? Compare the two ideas. Use a school-life example.
23. How is solids/liquids/gases related to classification.? Compare the two ideas. Use a real-world example.
24. How is classification. related to ■ Properties and uses of materials? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Properties and uses of materials related to solids/liquids/gases? Compare the two ideas. Explain in your own words.
26. How is solids/liquids/gases related to classification.? Compare the two ideas. Give a step-by-step response.
27. How is classification. related to ■ Properties and uses of materials? Compare the two ideas. Mention one common misconception.
28. How is ■ Properties and uses of materials related to solids/liquids/gases? Compare the two ideas. State one practical application.
29. How is solids/liquids/gases related to classification.? Compare the two ideas. Justify your response.
30. How is classification. related to ■ Properties and uses of materials? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Properties and uses of materials to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply solids/liquids/gases to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply classification. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Properties and uses of materials to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply solids/liquids/gases to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply classification. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Properties and uses of materials to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply solids/liquids/gases to a realistic situation. What would you do or calculate? State one practical application.
39. Apply classification. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Properties and uses of materials to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of solids/liquids/gases. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of classification.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Properties and uses of materials. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of solids/liquids/gases. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of classification.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Properties and uses of materials. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of solids/liquids/gases. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of classification.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Properties and uses of materials. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of solids/liquids/gases. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show classification.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties and uses of materials. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show solids/liquids/gases. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show classification.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties and uses of materials. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show solids/liquids/gases. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show classification.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Properties and uses of materials. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show solids/liquids/gases. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show classification.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 7. Temperature and its Measurement

Coverage: ■ Thermometers; temperature scales; heat vs temperature; measurement.

Understanding

1. What is ■ Thermometers, and what are its main features? Use a simple example.
2. What is temperature scales, and what are its main features? Use a school-life example.
3. What is heat vs temperature, and what are its main features? Use a real-world example.
4. What is measurement., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Thermometers, and what are its main features? Explain in your own words.
6. What is temperature scales, and what are its main features? Give a step-by-step response.
7. What is heat vs temperature, and what are its main features? Mention one common misconception.
8. What is measurement., and what are its main features? State one practical application.
9. What is ■ Thermometers, and what are its main features? Justify your response.
10. What is temperature scales, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to heat vs temperature and explain each. Use a simple example.
12. Give two examples related to measurement. and explain each. Use a school-life example.
13. Give two examples related to ■ Thermometers and explain each. Use a real-world example.
14. Give two examples related to temperature scales and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to heat vs temperature and explain each. Explain in your own words.
16. Give two examples related to measurement. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Thermometers and explain each. Mention one common misconception.
18. Give two examples related to temperature scales and explain each. State one practical application.
19. Give two examples related to heat vs temperature and explain each. Justify your response.
20. Give two examples related to measurement. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Thermometers related to temperature scales? Compare the two ideas. Use a simple example.
22. How is temperature scales related to heat vs temperature? Compare the two ideas. Use a school-life example.
23. How is heat vs temperature related to measurement.? Compare the two ideas. Use a real-world example.
24. How is measurement. related to ■ Thermometers? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Thermometers related to temperature scales? Compare the two ideas. Explain in your own words.
26. How is temperature scales related to heat vs temperature? Compare the two ideas. Give a step-by-step response.
27. How is heat vs temperature related to measurement.? Compare the two ideas. Mention one common misconception.
28. How is measurement. related to ■ Thermometers? Compare the two ideas. State one practical application.
29. How is ■ Thermometers related to temperature scales? Compare the two ideas. Justify your response.
30. How is temperature scales related to heat vs temperature? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply heat vs temperature to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply measurement. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Thermometers to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply temperature scales to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply heat vs temperature to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply measurement. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Thermometers to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply temperature scales to a realistic situation. What would you do or calculate? State one practical application.
39. Apply heat vs temperature to a realistic situation. What would you do or calculate? Justify your response.
40. Apply measurement. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Thermometers. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of temperature scales. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of heat vs temperature. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of measurement.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Thermometers. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of temperature scales. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of heat vs temperature. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of measurement.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Thermometers. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of temperature scales. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show heat vs temperature. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show measurement.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Thermometers. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show temperature scales. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show heat vs temperature. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show measurement.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Thermometers. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show temperature scales. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show heat vs temperature. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show measurement.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 8. Changes Around Us: States of Water

Coverage: NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus

Understanding

1. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a simple example.
2. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a school-life example.
3. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a real-world example.
4. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Show the idea with a diagram where appropriate.
5. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Explain in your own words.
6. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Give a step-by-step response.
7. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Mention one common misconception.
8. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? State one practical application.
9. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Justify your response.
10. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a simple example.
12. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a school-life example.
13. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a real-world example.
14. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Explain in your own words.
16. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Give a step-by-step response.
17. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Mention one common misconception.
18. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. State one practical application.
19. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Justify your response.
20. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Include the relevant rule or principle.

Comparison

21. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a simple example.
22. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a school-life example.

- 23.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a real-world example.
- 24.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Show the idea with a diagram where appropriate.
- 25.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Explain in your own words.
- 26.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Give a step-by-step response.
- 27.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Mention one common misconception.
- 28.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. State one practical application.
- 29.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Justify your response.
- 30.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Include the relevant rule or principle.

Application

- 31.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a simple example.
- 32.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a school-life example.
- 33.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a real-world example.
- 34.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
- 35.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Explain in your own words.
- 36.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Give a step-by-step response.
- 37.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Mention one common misconception.
- 38.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? State one practical application.
- 39.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Justify your response.
- 40.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

- 41.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a simple example.
- 42.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a school-life example.
- 43.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a real-world example.
- 44.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
- 45.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Explain in your own words.
- 46.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Give a step-by-step response.
- 47.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 9. Methods of Separation in Everyday Life

Coverage: ■ Sieving; filtration; sedimentation; decantation; evaporation; practical mixtures.

Understanding

1. What is ■ Sieving, and what are its main features? Use a simple example.
2. What is filtration, and what are its main features? Use a school-life example.
3. What is sedimentation, and what are its main features? Use a real-world example.
4. What is decantation, and what are its main features? Show the idea with a diagram where appropriate.
5. What is evaporation, and what are its main features? Explain in your own words.
6. What is practical mixtures., and what are its main features? Give a step-by-step response.
7. What is ■ Sieving, and what are its main features? Mention one common misconception.
8. What is filtration, and what are its main features? State one practical application.
9. What is sedimentation, and what are its main features? Justify your response.
10. What is decantation, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to evaporation and explain each. Use a simple example.
12. Give two examples related to practical mixtures. and explain each. Use a school-life example.
13. Give two examples related to ■ Sieving and explain each. Use a real-world example.
14. Give two examples related to filtration and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to sedimentation and explain each. Explain in your own words.
16. Give two examples related to decantation and explain each. Give a step-by-step response.
17. Give two examples related to evaporation and explain each. Mention one common misconception.
18. Give two examples related to practical mixtures. and explain each. State one practical application.
19. Give two examples related to ■ Sieving and explain each. Justify your response.
20. Give two examples related to filtration and explain each. Include the relevant rule or principle.

Comparison

21. How is sedimentation related to decantation? Compare the two ideas. Use a simple example.
22. How is decantation related to evaporation? Compare the two ideas. Use a school-life example.
23. How is evaporation related to practical mixtures.? Compare the two ideas. Use a real-world example.
24. How is practical mixtures. related to ■ Sieving? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Sieving related to filtration? Compare the two ideas. Explain in your own words.
26. How is filtration related to sedimentation? Compare the two ideas. Give a step-by-step response.
27. How is sedimentation related to decantation? Compare the two ideas. Mention one common misconception.
28. How is decantation related to evaporation? Compare the two ideas. State one practical application.
29. How is evaporation related to practical mixtures.? Compare the two ideas. Justify your response.
30. How is practical mixtures. related to ■ Sieving? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Sieving to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply filtration to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply sedimentation to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply decantation to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply evaporation to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply practical mixtures. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Sieving to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply filtration to a realistic situation. What would you do or calculate? State one practical application.
39. Apply sedimentation to a realistic situation. What would you do or calculate? Justify your response.
40. Apply decantation to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of evaporation. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of practical mixtures.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Sieving. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of filtration. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of sedimentation. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of decantation. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of evaporation. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of practical mixtures.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Sieving. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of filtration. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show sedimentation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show decantation. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show evaporation. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show practical mixtures.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Sieving. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show filtration. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show sedimentation. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show decantation. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show evaporation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show practical mixtures.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 10. Living Creatures: Exploring Characteristics

Coverage: ■ Growth; movement; respiration; reproduction; response; life processes.

Understanding

1. What is ■ Growth, and what are its main features? Use a simple example.
2. What is movement, and what are its main features? Use a school-life example.
3. What is respiration, and what are its main features? Use a real-world example.
4. What is reproduction, and what are its main features? Show the idea with a diagram where appropriate.
5. What is response, and what are its main features? Explain in your own words.
6. What is life processes., and what are its main features? Give a step-by-step response.
7. What is ■ Growth, and what are its main features? Mention one common misconception.
8. What is movement, and what are its main features? State one practical application.
9. What is respiration, and what are its main features? Justify your response.
10. What is reproduction, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to response and explain each. Use a simple example.
12. Give two examples related to life processes. and explain each. Use a school-life example.
13. Give two examples related to ■ Growth and explain each. Use a real-world example.
14. Give two examples related to movement and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to respiration and explain each. Explain in your own words.
16. Give two examples related to reproduction and explain each. Give a step-by-step response.
17. Give two examples related to response and explain each. Mention one common misconception.
18. Give two examples related to life processes. and explain each. State one practical application.
19. Give two examples related to ■ Growth and explain each. Justify your response.
20. Give two examples related to movement and explain each. Include the relevant rule or principle.

Comparison

21. How is respiration related to reproduction? Compare the two ideas. Use a simple example.
22. How is reproduction related to response? Compare the two ideas. Use a school-life example.
23. How is response related to life processes.? Compare the two ideas. Use a real-world example.
24. How is life processes. related to ■ Growth? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Growth related to movement? Compare the two ideas. Explain in your own words.
26. How is movement related to respiration? Compare the two ideas. Give a step-by-step response.
27. How is respiration related to reproduction? Compare the two ideas. Mention one common misconception.
28. How is reproduction related to response? Compare the two ideas. State one practical application.
29. How is response related to life processes.? Compare the two ideas. Justify your response.
30. How is life processes. related to ■ Growth? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Growth to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply movement to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply respiration to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply reproduction to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply response to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply life processes. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Growth to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply movement to a realistic situation. What would you do or calculate? State one practical application.
39. Apply respiration to a realistic situation. What would you do or calculate? Justify your response.
40. Apply reproduction to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of response. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of life processes.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Growth. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of movement. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of respiration. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of reproduction. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of response. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of life processes.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Growth. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of movement. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show respiration. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show reproduction. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show response. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show life processes.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Growth. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show movement. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show respiration. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show reproduction. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show response. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show life processes.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 11. Nature's Treasures

Coverage: ■ Air, water, soil, minerals; resources; conservation; sustainability.

Understanding

1. What is ■ Air, and what are its main features? Use a simple example.
2. What is water, and what are its main features? Use a school-life example.
3. What is soil, and what are its main features? Use a real-world example.
4. What is minerals, and what are its main features? Show the idea with a diagram where appropriate.
5. What is resources, and what are its main features? Explain in your own words.
6. What is conservation, and what are its main features? Give a step-by-step response.
7. What is sustainability., and what are its main features? Mention one common misconception.
8. What is ■ Air, and what are its main features? State one practical application.
9. What is water, and what are its main features? Justify your response.
10. What is soil, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to minerals and explain each. Use a simple example.
12. Give two examples related to resources and explain each. Use a school-life example.
13. Give two examples related to conservation and explain each. Use a real-world example.
14. Give two examples related to sustainability. and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to ■ Air and explain each. Explain in your own words.
16. Give two examples related to water and explain each. Give a step-by-step response.
17. Give two examples related to soil and explain each. Mention one common misconception.
18. Give two examples related to minerals and explain each. State one practical application.
19. Give two examples related to resources and explain each. Justify your response.
20. Give two examples related to conservation and explain each. Include the relevant rule or principle.

Comparison

21. How is sustainability. related to ■ Air? Compare the two ideas. Use a simple example.
22. How is ■ Air related to water? Compare the two ideas. Use a school-life example.
23. How is water related to soil? Compare the two ideas. Use a real-world example.
24. How is soil related to minerals? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is minerals related to resources? Compare the two ideas. Explain in your own words.
26. How is resources related to conservation? Compare the two ideas. Give a step-by-step response.
27. How is conservation related to sustainability.? Compare the two ideas. Mention one common misconception.
28. How is sustainability. related to ■ Air? Compare the two ideas. State one practical application.
29. How is ■ Air related to water? Compare the two ideas. Justify your response.
30. How is water related to soil? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply soil to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply minerals to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply resources to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply conservation to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply sustainability. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Air to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply water to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply soil to a realistic situation. What would you do or calculate? State one practical application.
39. Apply minerals to a realistic situation. What would you do or calculate? Justify your response.
40. Apply resources to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of conservation. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of sustainability.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Air. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of water. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of soil. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of minerals. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of resources. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of conservation. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of sustainability.. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of ■ Air. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show water. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show soil. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show minerals. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show resources. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show conservation. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show sustainability.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Air. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show water. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show soil. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show minerals. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 12. Beyond Earth

Coverage: ■ Solar system; stars; planets; Moon; space exploration and observation.

Understanding

1. What is ■ Solar system, and what are its main features? Use a simple example.
2. What is stars, and what are its main features? Use a school-life example.
3. What is planets, and what are its main features? Use a real-world example.
4. What is Moon, and what are its main features? Show the idea with a diagram where appropriate.
5. What is space exploration and observation., and what are its main features? Explain in your own words.
6. What is ■ Solar system, and what are its main features? Give a step-by-step response.
7. What is stars, and what are its main features? Mention one common misconception.
8. What is planets, and what are its main features? State one practical application.
9. What is Moon, and what are its main features? Justify your response.
10. What is space exploration and observation., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Solar system and explain each. Use a simple example.
12. Give two examples related to stars and explain each. Use a school-life example.
13. Give two examples related to planets and explain each. Use a real-world example.
14. Give two examples related to Moon and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to space exploration and observation. and explain each. Explain in your own words.
16. Give two examples related to ■ Solar system and explain each. Give a step-by-step response.
17. Give two examples related to stars and explain each. Mention one common misconception.
18. Give two examples related to planets and explain each. State one practical application.
19. Give two examples related to Moon and explain each. Justify your response.
20. Give two examples related to space exploration and observation. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Solar system related to stars? Compare the two ideas. Use a simple example.
22. How is stars related to planets? Compare the two ideas. Use a school-life example.
23. How is planets related to Moon? Compare the two ideas. Use a real-world example.
24. How is Moon related to space exploration and observation.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is space exploration and observation. related to ■ Solar system? Compare the two ideas. Explain in your own words.
26. How is ■ Solar system related to stars? Compare the two ideas. Give a step-by-step response.
27. How is stars related to planets? Compare the two ideas. Mention one common misconception.
28. How is planets related to Moon? Compare the two ideas. State one practical application.
29. How is Moon related to space exploration and observation.? Compare the two ideas. Justify your response.
30. How is space exploration and observation. related to ■ Solar system? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Solar system to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply stars to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply planets to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply Moon to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply space exploration and observation. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Solar system to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply stars to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply planets to a realistic situation. What would you do or calculate? State one practical application.
39. Apply Moon to a realistic situation. What would you do or calculate? Justify your response.
40. Apply space exploration and observation. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Solar system. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of stars. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of planets. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of Moon. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of space exploration and observation.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Solar system. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of stars. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of planets. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of Moon. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of space exploration and observation.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Solar system. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show stars. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show planets. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show Moon. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show space exploration and observation.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Solar system. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show stars. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show planets. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show Moon. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show space exploration and observation.. Include the relevant rule or principle.